Exponentials & Logarithms

ANSWER KEY

Section 1: The Natural Exponential & Logarithm

1 Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \dfrac{1 + \ln 5}{2}$ ✓

b. $x = \dfrac{\ln 5 - 1}{2}$

c. $x = \ln\left(\dfrac{5}{2}\right) + 1$

d. $x = \dfrac{\ln 6}{2}$

Take natural logarithms of both sides: $2x - 1 = \ln 5 \Rightarrow 2x = 1 + \ln 5 \Rightarrow x = \frac{1+\ln 5}{2}$.

2 Solve $\ln(3x + 2) - \ln(x) = \ln 4$ for $x > 0$.

a. $x = 2$ ✓

b. $x = 1$

c. $x = 4$

d. $x = 1/2$

$\ln\left(\frac{3x+2}{x}\right) = \ln 4 \Rightarrow \frac{3x+2}{x} = 4 \Rightarrow 3x + 2 = 4x \Rightarrow x = 2$.

3 Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.

a. $x^3 + 2x$ ✓

b. $3x + 2x = 5x$

c. $x^3 + e^{2x}$

d. $3x^2 + 2$

$e^{3\ln x} = e^{\ln(x^3)} = x^3$. $\ln(e^{2x}) = 2x$. Sum is $x^3 + 2x$.

Section 2: Exponential Equations with Different Bases

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4 Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.

a. $x = \dfrac{\ln 2}{\ln 3 - \ln 2}$ ✓

b. $x = \dfrac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \dfrac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \dfrac{1}{\ln(3/2)}$

$\ln(2^{x+1}) = \ln(3^x) \Rightarrow (x+1)\ln 2 = x\ln 3 \Rightarrow x\ln 2 + \ln 2 = x\ln 3 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

5 Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$ ✓

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

Multiply by e^x: $(e^x)^2 - e^x - 6 = 0 \Rightarrow (e^x - 3)(e^x + 2) = 0$. Since $e^x > 0$, $e^x = 3 \Rightarrow x = \ln 3$.

6 Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10 ✓

b. 0.852

c. 1.21

d. 0.945

$(2x-1)\ln 5 = \ln 7 \Rightarrow 2x - 1 = \frac{\ln 7}{\ln 5} = \frac{1.9459}{1.6094} \approx 1.209 \Rightarrow 2x \approx 2.209 \Rightarrow x \approx 1.10$.

Section 3: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

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7 The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. **Gradient** $= \ln b$**, Intercept** $= \ln a$ ✓

b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

Take logarithms: $\ln y = \ln(a \cdot b^x) = \ln a + x \ln b = (\ln b)x + \ln a$. This matches $Y = mX + c$ with gradient $\ln b$ and intercept $\ln a$.

8 The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.

a. $k = 4, n = 2.5$ ✓

b. $k = 2.5, n = 4$

c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

$\ln y = n \ln x + \ln k$. Gradient $n = 2.5$. Intercept $\ln k = 1.386 \Rightarrow k = e^{1.386} = 4$.

9 Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.

a. $b = e^{0.8} \approx 2.226$ ✓

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

Gradient $m = \ln b = \frac{5.6 - 3.2}{5 - 2} = \frac{2.4}{3} = 0.8$. Thus $b = e^{0.8} \approx 2.226$.

Section 4: Exponential Growth and Decay Modeling

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10 The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years ✓

b. 20.0 years

c. 10.0 years

d. 15.5 years

$M(t) = 40 \Rightarrow 80e^{-0.05t} = 40 \Rightarrow e^{-0.05t} = 0.5 \Rightarrow -0.05t = -\ln 2 \Rightarrow t = \frac{\ln 2}{0.05} = 20\ln 2 \approx 13.86$ years.

11 A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \dfrac{\ln 4}{3}$ ✓

b. $k = \dfrac{\ln 2}{3}$

c. $k = \dfrac{4}{3}$

d. $k = \ln\left(\dfrac{4}{3}\right)$

$500e^{3k} = 2000 \Rightarrow e^{3k} = 4 \Rightarrow 3k = \ln 4 \Rightarrow k = \frac{\ln 4}{3}$.

12 A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min ✓

b. 15.2 min

c. 10.5 min

d. 18.0 min

$45 = 20 + 70e^{-0.08t} \Rightarrow 25 = 70e^{-0.08t} \Rightarrow e^{-0.08t} = 25/70 = 5/14 \Rightarrow -0.08t = \ln(5/14) \approx -1.0296 \Rightarrow t \approx 1.0296/$

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